

Supplementary Information: Unraveling Electronic Trends in O* and OH* Surface Adsorption in the MO₂ Transition-Metal Oxide Series

Benjamin M. Comer,^{†,‡} Jiang Li,^{†,‡} Frank Abild-Pedersen,[‡] Michal Bajdich,[‡] and
Kirsten T. Winther^{*,‡}

[†] *SUNCAT Center for Interface Science and Catalysis, Department of Chemical
Engineering, Stanford University, Stanford, CA 94305, USA*

[‡] *SUNCAT Center for Interface Science and Catalysis, SLAC National Accelerator
Laboratory, Menlo Park, CA 94025, USA*

E-mail: winther@slac.stanford.edu

Hubbard U values

Table S1: Hubbard U values used for bulk and surface DFT calculations for 3d transition metal oxides. Values are adapted from Materials Project,² Ref.,³ and Ref.¹

Element	Hubbard U Value
Ti	3.00 ¹
V	3.25 ²
Cr	3.5 ¹
Mn	3.75 ¹
Fe	4.3 ³
Co	3.32 ²
Ni	6.45 ²
Cu	3.0 ¹

Magnetic structure

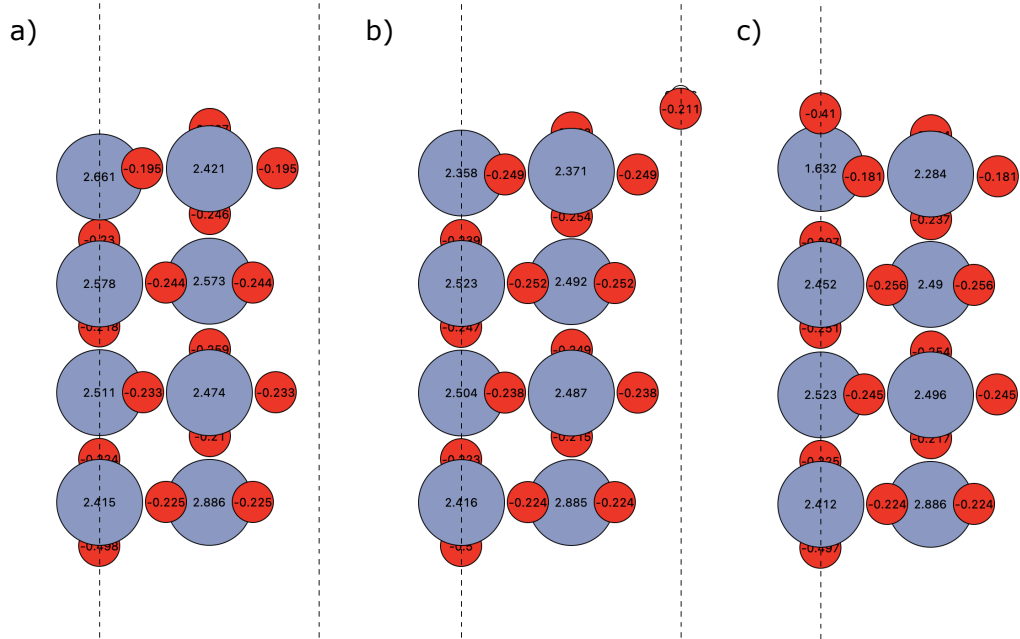


Figure S1: A comparison of the magnetic moments for (110) surfaces. The numbers on each atom show the magnetic moment in Bohr magnetons. The a) clean surface b) OH* adsorbed, and c) O* adsorbed surfaces show identical magnetic moments on the bottom half of the slab. The only significant changes in magnetic moment are near the active site.

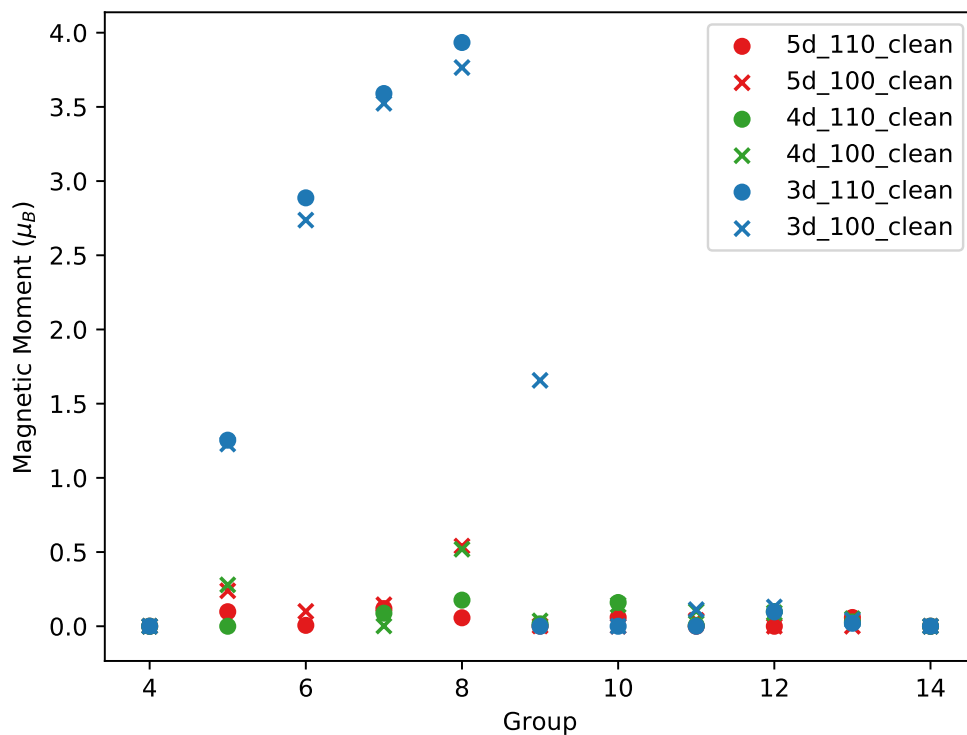


Figure S2: The magnetic moments on the active metal site for clean surface (100) and (110) rutile surfaces, values plotted in Bohr magnetons. While all 4d and 5d metals have low spin configurations, the early 3d metals have high spin. The spin magnitude increases monotonically with the number of d-electrons and reaches a maximum for Fe, followed by a switch to low-spin configuration for Co.

Theoretical OER overpotential

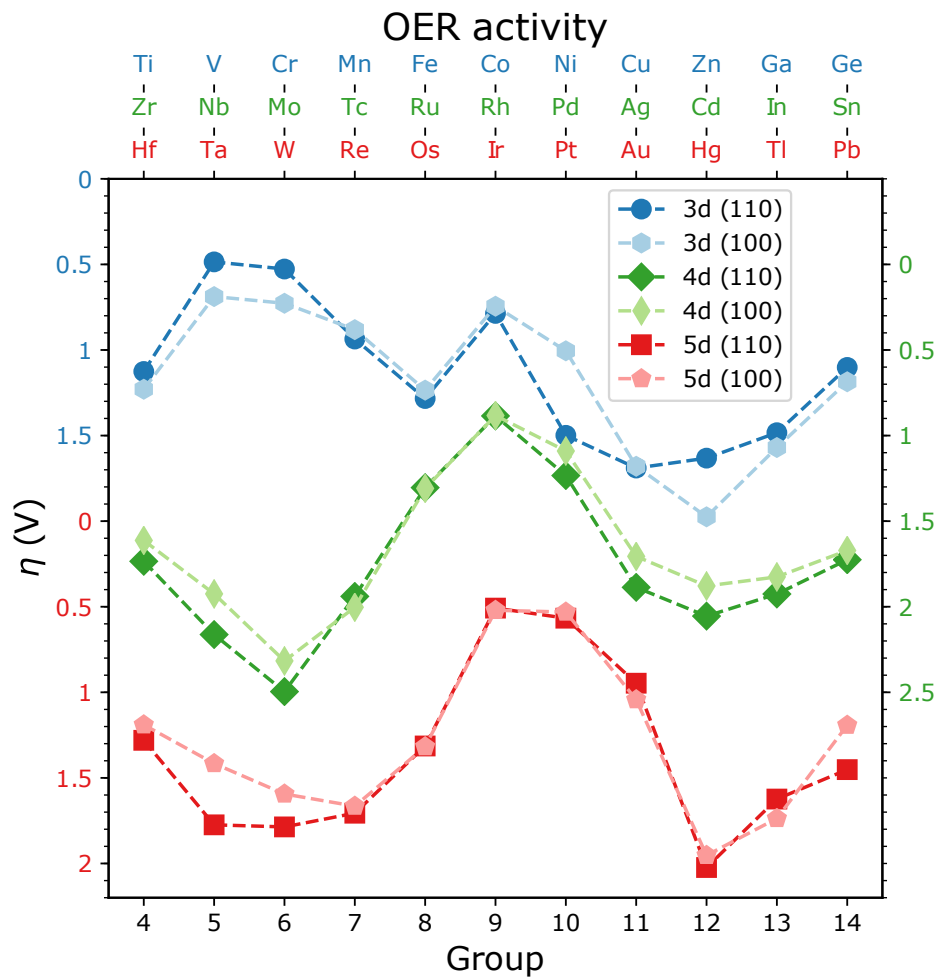
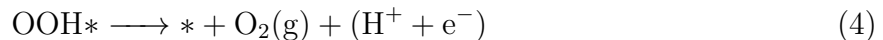
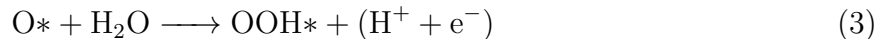
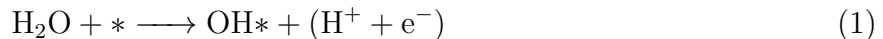


Figure S3: The OER overpotential of rutile oxide surfaces with respect to the column on the periodic table. The free energy corrections used are described below.

The calculated Oxygen Evolution Reaction (OER) overpotentials for the MO₂ Rutile oxides are shown in Figure S3. We assume a 4-electron mechanism through the OOH*

intermediate:



We approximate the OOH adsorption energy from the scaling relation: $\Delta G_{\text{OOH}} = 0.84\Delta G_{\text{OH}} + 3.14\text{eV}$ taken from Ref.¹ The OER overpotential is then defined from the difference between the maximum free energy step (Equations 1 - 4 above) and the theoretical water splitting potential; $\mu_{\text{OER}} = \max[\Delta G_1, \Delta G_2, \Delta G_3, \Delta G_4]/e + 1.23\text{V}$. Free energy corrections are applied to DFT O^* and OH^* adsorption energies to obtain Gibbs free adsorption energies, corresponding to the equation: $\Delta G_i = \Delta E_i + \Delta \text{ZPE}_i - T\Delta S_i$. Correction values are based on vibrational frequency calculations and were adapted from previous work¹ tabulated in Table S2.

Table S2: The free energy corrections used to generate Figure S3. All values presented are in units of eV. It should be noted that these corrections were not applied to the values in the main text. These corrections were adapted from previous work.¹

Species	ZPE	H _{vib}	TS	ZPE + H _{vib} - TS
H ₂ O	0.560	0.103	0.675	-0.012
H ₂	0.268	0.091	0.408	-0.049
O*	0.065	0.038	0.080	0.023
OH*	0.344	0.051	0.085	0.310

Adsorption energy trends in +4 transition metal oxides

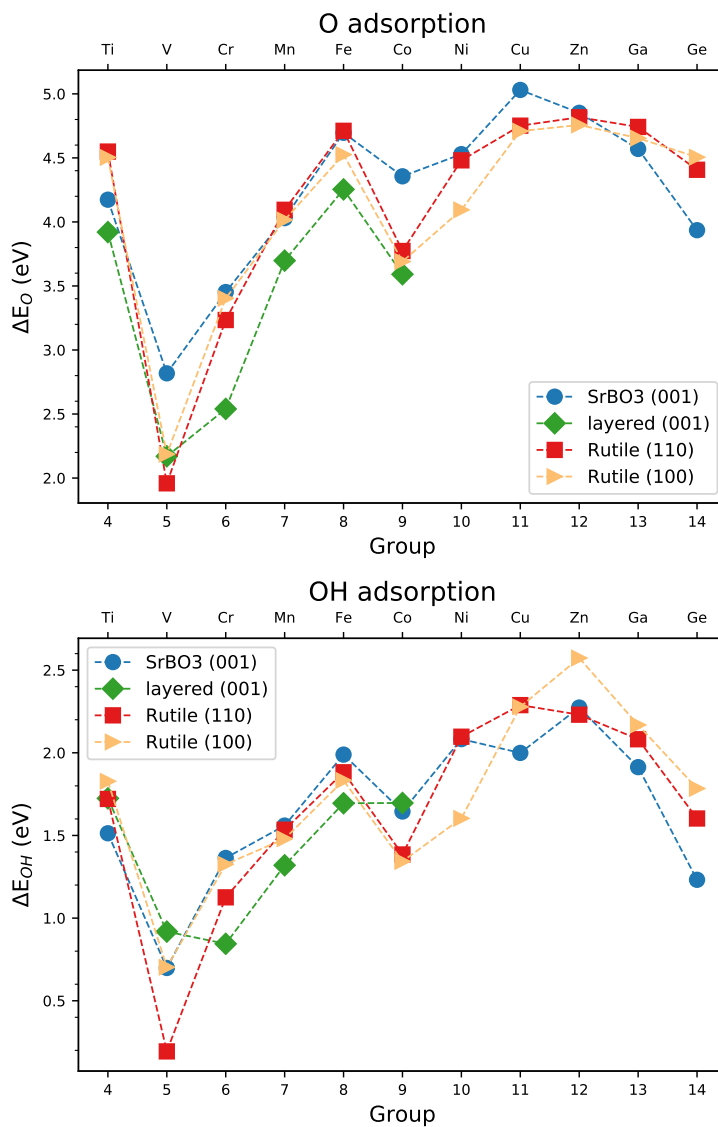


Figure S4: The adsorption energy of O and OH on different 3d transition metal (+4) oxide structures, including layered MO_2 , perovskite SrMO_3 , and MO_2 rutile (110) and (100) surfaces. Layered oxide data was adopted from Ref,¹ and Perovskite data was computed with the same calculational settings as this work, except only the adsorbate were allowed to relax.

Supplementary ICOHP results

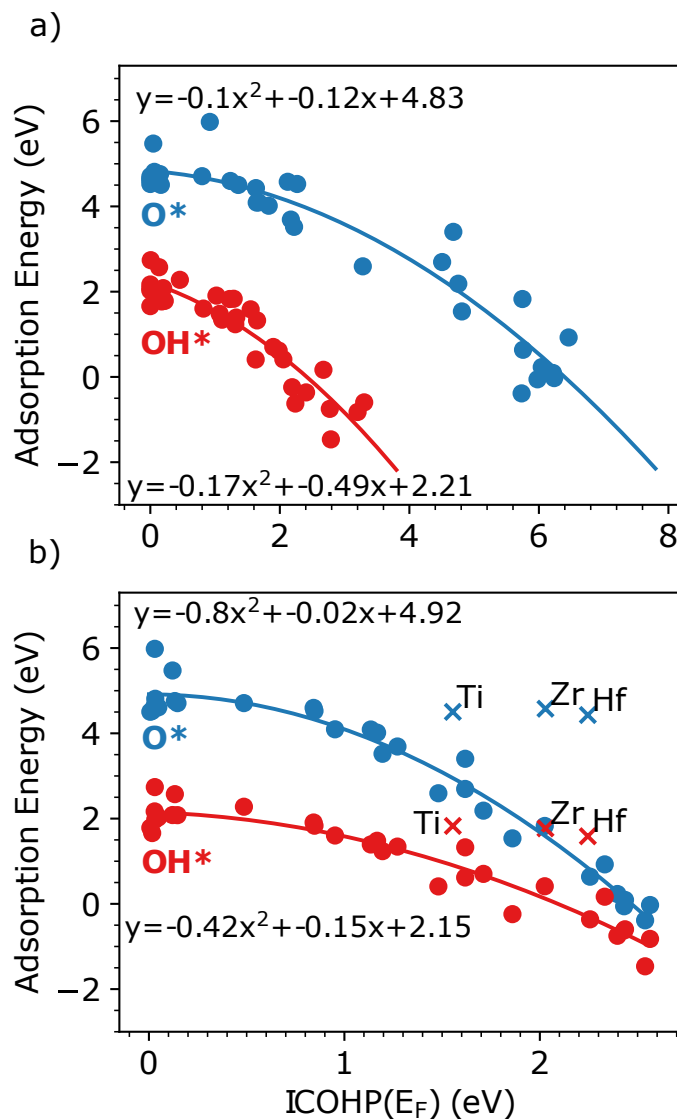


Figure S5: Quadratic model between 100 ICOHP and adsorption energy using (a) surface ICOHP and the RMSE/MAE are 0.56/0.42 eV and 0.42/0.33 eV for O^* and OH^* respectively (b) bulk ICOHP the RMSE/MAE are 0.36/0.28 eV and 0.32/0.24 eV for O^* and OH^* respectively. The x points are not included in b) due to the mismatch in behaviour between the surface and bulk for the group 4 metals.

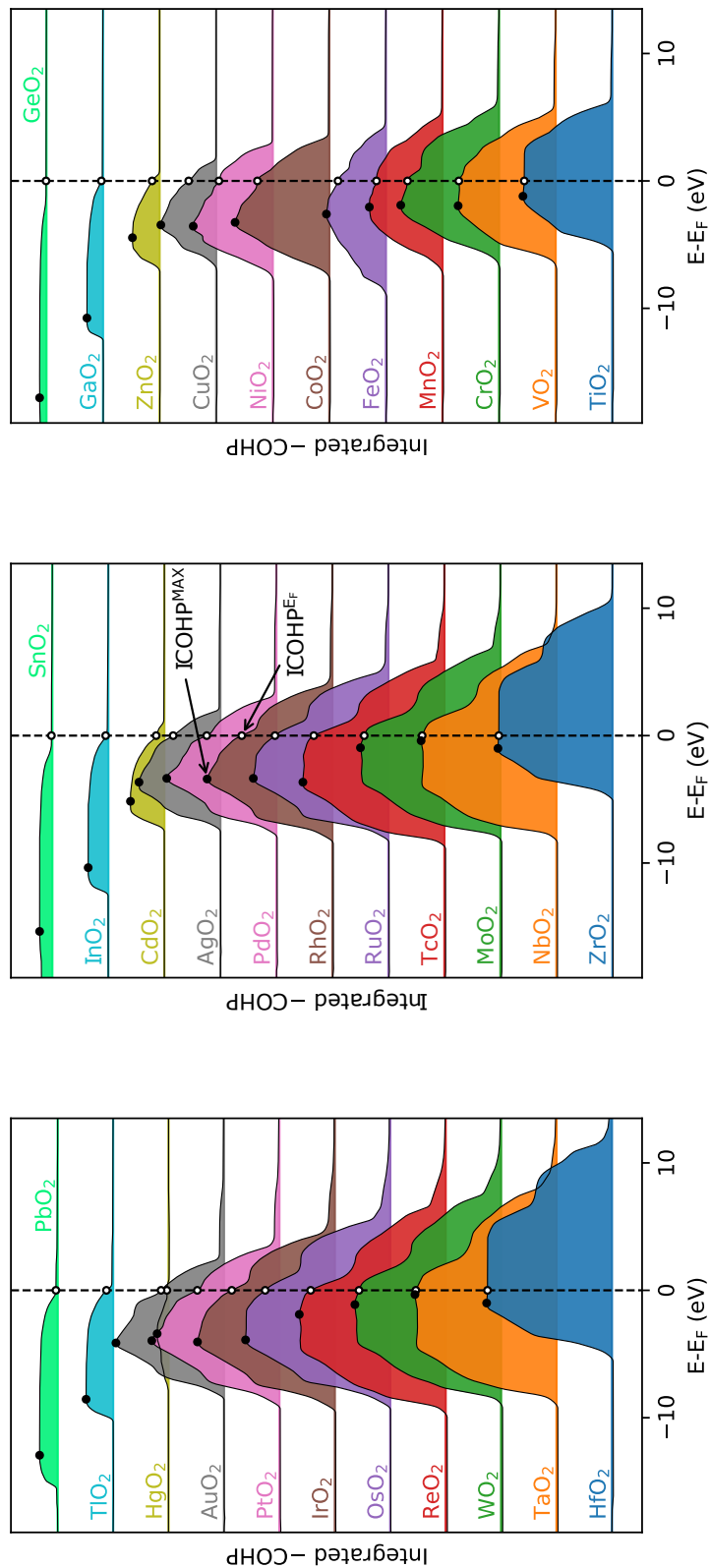


Figure S6: The integrated COHP for the bulk rutile 3d, 4d, and 5d metal oxides. 4d metals are in Figure 4a) in the main text. The black points represent the ICOHP(MAX) and the white filled points represent ICOHP(E_F). S8

Tabulated adsorption energies

Table S3: The adsorption energies obtained for O* and OH* adsorption on rutile 110 and 100 surfaces. The data and structures are stored on Catalysis Hub at <https://www.catalysis-hub.org/publications/ComerUnraveling2022>

Surface	Facet	Reaction	ΔE_{ads}	ΔG_{ads}
CdO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	5.02	5.01
CdO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.26	2.56
CdO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.71	4.7
CdO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.08	2.38
WO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	-0.75	-0.76
WO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-1.45	-1.15
WO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	-0.03	-0.04
WO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.82	-0.52
PdO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.12	4.11
PdO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	0.88	1.18
PdO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	3.52	3.51
PdO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.24	1.54
NbO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	-0.25	-0.26
NbO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-1.03	-0.73
NbO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	0.23	0.22
NbO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.75	-0.45
CuO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.75	4.74
CuO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.29	2.59
CuO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.71	4.7
CuO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.28	2.58
HgO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	5.45	5.44
HgO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.81	3.11

HgO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	5.98	5.97
HgO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.74	3.04
ZnO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.82	4.81
ZnO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.23	2.53
ZnO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.76	4.75
ZnO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.57	2.87
PtO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	2.73	2.72
PtO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.5	0.8
PtO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	2.59	2.58
PtO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.41	0.71
AgO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.78	4.77
AgO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.09	2.39
AgO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.6	4.59
AgO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.91	2.21
GaO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.74	4.73
GaO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.08	2.38
GaO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.65	4.64
GaO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.17	2.47
OsO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	0.39	0.38
OsO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-0.66	-0.36
OsO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	0.64	0.63
OsO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-0.36	-0.06
VO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	1.96	1.95
VO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.19	0.49
VO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	2.19	2.18
VO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.7	1.0

CrO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	3.23	3.22
CrO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.13	1.43
CrO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	3.4	3.39
CrO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.33	1.63
TaO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	-1.07	-1.08
TaO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-1.85	-1.55
TaO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	-0.39	-0.4
TaO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-1.46	-1.16
FeO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.71	4.7
FeO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.88	2.18
FeO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.53	4.52
FeO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.83	2.13
RhO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	2.54	2.53
RhO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.46	0.76
RhO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	2.7	2.69
RhO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	0.62	0.92
InO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.85	4.84
InO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.13	2.43
InO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.61	4.6
InO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.03	2.33
ReO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	-0.5	-0.51
ReO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-1.25	-0.95
ReO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	0.09	0.08
ReO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-0.6	-0.3
IrO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	1.46	1.45
IrO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	-0.34	-0.04

IrO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	1.54	1.53
IrO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.24	0.06
GeO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.41	4.4
GeO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.6	1.9
GeO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.51	4.5
GeO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.78	2.08
AuO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.11	4.1
AuO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.51	1.81
AuO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.09	4.08
AuO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.39	1.69
RuO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	1.63	1.62
RuO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	0.17	0.47
RuO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	1.83	1.82
RuO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	0.41	0.71
PbO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	5.11	5.1
PbO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.0	2.3
PbO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.81	4.8
PbO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.98	2.28
HfO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.47	4.46
HfO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.54	1.84
HfO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.43	4.42
HfO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.59	1.89
TcO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	0.54	0.53
TcO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.36	-0.06
TcO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	0.93	0.92
TcO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	0.17	0.47

CoO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	3.77	3.76
CoO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.38	1.68
CoO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	3.69	3.68
CoO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.34	1.64
MnO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.09	4.08
MnO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.53	1.83
MnO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.02	4.01
MnO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.48	1.78
ZrO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.64	4.63
ZrO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.72	2.02
ZrO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.58	4.57
ZrO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.77	2.07
MoO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	-0.28	-0.29
MoO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.67	-0.37
MoO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	-0.06	-0.07
MoO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	-0.62	-0.32
NiO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.48	4.47
NiO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.1	2.4
NiO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.09	4.08
NiO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.6	1.9
TiO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.55	4.54
TiO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.72	2.02
TiO ₂	100	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	4.5	4.49
TiO ₂	100	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	1.83	2.13
TlO ₂	110	$\text{H}_2\text{O(g)} - \text{H}_2\text{(g)} + * \rightarrow \text{O}^*$	5.08	5.07
TlO ₂	110	$\text{H}_2\text{O(g)} - 0.5\text{H}_2\text{(g)} + * \rightarrow \text{HO}^*$	2.41	2.71

TiO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	5.47	5.46
TiO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	2.08	2.38
SnO ₂	110	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.43	4.42
SnO ₂	110	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.4	1.7
SnO ₂	100	$\text{H}_2\text{O}(\text{g}) - \text{H}_2(\text{g}) + * \rightarrow \text{O}^*$	4.53	4.52
SnO ₂	100	$\text{H}_2\text{O}(\text{g}) - 0.5\text{H}_2(\text{g}) + * \rightarrow \text{HO}^*$	1.66	1.96

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